

L-11 [Chapter 3.2, 3.3]

□ Rounding Error:

Max
The Error of Central Differentiation
= Truncation Error + Rounding Error

$$= \underbrace{\left| \frac{f'''(\xi) h^2}{3!} \right|}_{\text{error} \propto h^2} + \underbrace{\epsilon_m \cdot \left| \frac{f(x_0+h) - f(x_0-h)}{2h} \right|}_{\text{error} \propto \frac{1}{h}}$$

Machine epsilon

error $\propto h^2$
if, $h \downarrow$ Error $\downarrow$
 $h \uparrow$ Error $\uparrow$

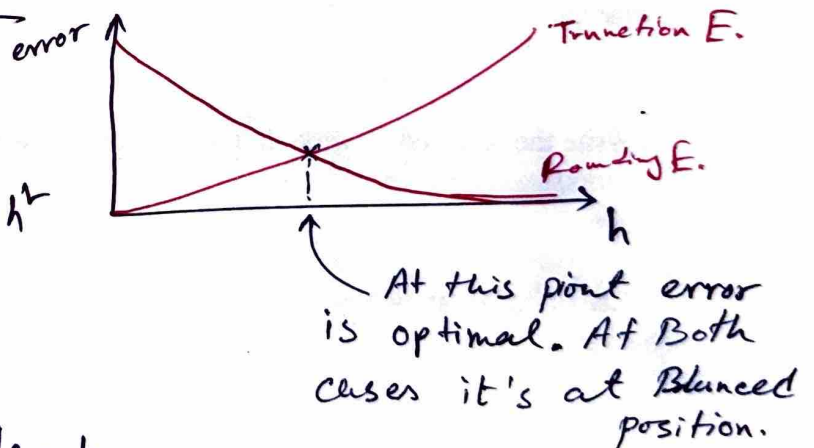
error $\propto \frac{1}{h}$
if $h \downarrow$ error $\uparrow$
 $h \uparrow$ error $\downarrow$

Let's plot graph

Observations

Central dif. $\frac{f''(\xi) h^2}{3!} \Rightarrow$ error order h^2
 $\therefore O(h^2)$

forward dif. $\frac{f'(\xi) h}{2} \Rightarrow$ error order h
 $\therefore O(h)$



Richardson Extrapolation [applicable for only central dif error]

$$D_h = \frac{f(x+h) - f(x-h)}{2h}$$

Using Taylor's series. [upto 5th order approximation]

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2!} + \frac{f'''(x)h^3}{3!} + \frac{f^{(4)}(x)h^4}{4!} + \frac{f^{(5)}(x)h^5}{5!} + O(h^6)$$

$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)h^2}{2!} - \frac{f'''(x)h^3}{3!} + \frac{f^{(4)}(x)h^4}{4!} - \frac{f^{(5)}(x)h^5}{5!} + O(h^6)$$

Now, $D_h = \frac{f(x+h) - f(x-h)}{2h}$

$$= \frac{1}{2h} \left(2f'(x)h + \frac{2 \cdot f'''(x)h^3}{3!} + \frac{2f^{(5)}(x)h^5}{5!} + O(h^7) \right)$$

$$\therefore D_h = \underbrace{f'(x)}_{\text{differentiation term}} + \underbrace{\frac{f'''(x)h^2}{3!} + \frac{f^{(5)}(x)h^4}{5!} + O(h^6)}_{\text{Truncation error}}$$

Goal is to Remove $\rightarrow \boxed{\frac{f'''(x)h^2}{3!}}$

Cause: this term is impacting more effect to Error.

Even though lowering $h \downarrow$, error is not lowering that **MUCH**.

$$D_h = f'(x) + \frac{f'''(x)}{3!} h^2 + \frac{f^{(5)}(x)}{5!} h^4 + O(h^6) \dots \textcircled{i}$$

$$D_{h/2} = f'(x) + \frac{f'''(x)}{3!} \cdot \frac{h^2}{4} + \frac{f^{(5)}(x)}{5!} \cdot \frac{h^4}{16} + O(h^6) \dots \textcircled{ii}$$

$$4 \cdot \textcircled{ii} - \textcircled{i} \Rightarrow$$

$$\Rightarrow 4 \cdot D_{h/2} - D_h = 3f'(x) + 0 + \left(\frac{1}{4} - 1\right) \frac{f^{(5)}(x) h^4}{5!} + O(h^6)$$

$$\frac{4D_{h/2} - D_h}{3} = f'(x) - \frac{1}{4} \frac{f^{(5)}(x)}{5!} h^4 + O(h^6) \quad \left. \vphantom{\frac{4D_{h/2} - D_h}{3}} \right] \swarrow D_h^{(1)}$$

Let's do for $D_h^{(2)}$

$$D_h^{(1)} = f'(x) - \frac{1}{4} \frac{f^{(5)}(x)}{5!} h^4 + O(h^6)$$

$$D_{h/2}^{(1)} = f'(x) - \frac{1}{4} \frac{f^{(5)}(x)}{5!} \frac{h^4}{16} + O(h^6)$$

$$16 D_{h/2}^{(1)} - D_h^{(1)} = 15 f'(x) + O(h^6)$$

$$\frac{16 D_{h/2}^{(1)} - D_h^{(1)}}{15} = f'(x) + O(h^6) \quad \left. \vphantom{\frac{16 D_{h/2}^{(1)} - D_h^{(1)}}{15}} \right] \rightarrow D_h^{(2)}$$